PORTLAND PUBLIC SCHOOLS
Office of School Modernization
Memorandum to File
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MEMORANDUM

Date: September 5, 2025

Subject: Structural Evaluation Process for HS Modernizations

From: Stormy Shanks - Senior Director, OSM

To: PPS Leadership

This memo is to provide clarity surrounding the structural system evaluation process for current PPS High School Modernization projects. All three design teams for the Cleveland, Ida B. Wells, and Jefferson High School Modernizations are recommending a hybrid structural approach which will include the use of Mass Timber, Steel, and Concrete or CMU.

Until recent years, most large scale building projects have been typically designed and constructed using steel and concrete/CMU as the basis for structural design. Technology has advanced considerably to make the use of wood more sustainable in large-scale construction. The district has a responsibility to ensure the highest and best use of each dollar allocated to our improvement projects as well as ensure the best learning environments for students. As technology advances, it becomes increasingly necessary to examine and stay updated on best practices and adjust accordingly.

The choice of a structural system to be used on a large-scale building has many effects both short and long term and requires deep analysis. In cost analysis provided by Skanska, the Contractor for the Cleveland Modernization, PPS learned that Mass Timber has become relatively cost neutral compared to the more typical structural systems mentioned above due to the faster assembly time during construction. Given that cost neutrality, our projects' structural systems needed to be analyzed using other criteria to determine whether appropriate. Cleveland is a bit farther along in their design schedule and moved forward with Mass Timber as the assumed basis of design. During that coordination process, our teams began fielding many questions about Mass Timber primarily from our Facilities and Operations stakeholders. These questions centered around cleaning, durability, flexibility, environmental benefits, etc. It was determined by OSM project teams that further analysis was necessary to clarify these issues for our stakeholders.

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The analysis strategies used are partially detailed in a memo provided to OSM by one of the design teams, Bora Architects. Their analysis was completed under the IBW and Jefferson HS Modernizations scope. However, the analysis compiled applies to all three modernizations.

The design team timed the structural analysis to coincide with the issuance of Schematic Design Phase documentation. It is typical that projects do various forms of analysis of structures and systems during Schematic Design and then finalize decisions based on that analysis during the Design Development phase of the construction documents. When Jefferson's Schematic Design Drawings are approved, all three modernization projects will then be in Design Development. On 9/3/25, there was a Mass Timber Summit, led by Bora, which included OSM, Facilities and Operations staff, and PPS Leadership. Bora explained and demonstrated the tradeoffs and advantages to using Mass Timber, why the design team is making its recommendation, and to answer final questions staff had.

Following the Mass Timber Summit, the A/E teams are moving forward into Design Development to further coordinate construction drawings around the recommendation to use Mass Timber.

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